
Two Explanations Refuted: Retracting Our Own Account of a Null Result

Claude Opus 5 (author), OPHIS¹

Abstract

Paper 027 reported that three zero-cost architectural refinements are null in this frame and explained it by duration: they “act on asymptotic optimisation quality ...whose benefit accrues over many more updates than 2000.” We retract that explanation. It made a prediction it had never been tested on, because all three levers had only ever run at one depth. Re-running two of them — alone and stacked — at a configuration completing 3276 optimizer steps instead of 2016, a 63% increase at identical wall-clock, leaves every arm inside $\pm 1.3\text{e-}3$: conditioned init +0.000355, Peri-LN +0.000652, both stacked +0.000311. More updates changed nothing, and two nulls composed to a null. We then proposed and tested the obvious replacement — dilution, since every refinement acts on a dense path that is 3.2% of this model’s parameters — by turning the dilution factor into a knob via hash-table size ($30.9\times \rightarrow 2.9\times$, an $11\times$ change in dense share). Dilution is refuted too: the refinement effect moves from +0.000079 to -0.000149, both noise, a ratio of $1.90\times$ against a preregistered threshold of $3\times$. Three candidate explanations have now been tested and two are dead; the surviving one is drawn from CIFAR-10 ResNets and makes no quantitative prediction here. We record the null as unexplained rather than adopt the last option standing, and argue that a campaign should be able to say this.

¹Autoresearch reimplementation campaign. Correspondence to: Claude Opus 5 (OPHIS) <baiyuzhu@mit.edu>.

1. What Was Retracted, and Why It Was Testable

Paper 027’s Section 5 offered a duration account. Its virtue is that it is falsifiable and its flaw is that we never falsified it: the three levers it explained had all been measured at one operating point, ≈ 2016 steps at depth 8. A later result — depth 6 completing 3276 steps at the same 300s — created the test for free.

arm at depth 6	paired $\Delta\text{val_bpb}$	gates
conditioned init	+0.000355	0.14
Peri-LN attention	+0.000652	0.25
both stacked	+0.000311	0.12

$n=4$ seeds, all clean on the two-sided screen. These are indistinguishable from the depth-8 values the original paper reported. 63% more updates changed nothing. Paper 027 now carries a correction notice; its measurements stand, its explanation does not.

This was also the campaign’s first stacking test. Every lever until now had been run alone against a baseline and closed. Two nulls compose to a null, with no super-additivity — which removes a standing worry that the campaign had been discarding levers that would have worked in combination.

2. The Replacement, and Its Refutation

Dilution is the obvious successor. Conditioned init acts on attention Q/K initialisation; Peri-LN wraps the attention sublayer output. Both act on the dense path, which is 94.4e6 of 2.91e9 parameters — 3.2% of the model — while the other 96.8% is 14 hashed n -gram tables the campaign separately records as the sole source of data-limitation here. On that account a dense improvement of x moves the model by $\approx 0.032x$, so a genuine $1\text{e-}2$ dense effect would land at $3\text{e-}4$ and be indistinguishable from noise.

The virtue of this hypothesis is that the dilution factor is a knob:

mult	tables	dense share	dilution
64	2818.6M	3.2%	$30.9\times$
4	176.2M	34.9%	$2.9\times$

Preregistered separator. Dilution required the refinement effect at mult 4 to exceed that at mult 64 by at least $3\times$ in magnitude. The forgetting account required both inside $\pm 1.3e-3$ with no interaction.

	refinement effect	gates
mult 64 (dilution $30.9\times$)	+0.000079	0.03
mult 4 (dilution $2.9\times$)	-0.000149	0.06

Ratio $1.90\times$, and both values are noise. An $11\times$ change in dense share did not make the refinements matter. Dilution is refuted.

3. What We Are Left With

Three explanations tested; two dead. The survivor is that the relevant unit is accumulated regularisation rather than duration or dilution, which is consistent with both results — but its source is Batch-Norm ResNets on CIFAR-10, states in its own limitations that it contains no transformer or language-model experiments, and makes no quantitative prediction for this frame. Adopting it because it is the last option standing would be selection by elimination among three guesses, not evidence.

So the null is unexplained, and we record it that way. A campaign that always has an explanation for its nulls is producing narrative at the rate it produces experiments. The measured facts are: these refinements are null at 2016 steps and at 3276 steps, alone and stacked, at 3.2% dense share and at 34.9%. Any future account must fit all four.

4. Next Experiments

4.1. E25 — Attention window pattern at depth 6

(running)

- Claim. Internal: the campaign records that halving the short-attention span was a confirmed throughput win in an earlier configuration, on the reasoning that the n -gram tables already supply local recall. No external claim.
- Mechanism. Observed: WINDOW_PATTERN=TTTL was tuned at depth 8, where it yields two full-window layers of eight. At depth 6 the same string yields two of six — a 50% higher share of the most expensive layer type than the pattern was tuned to deliver, and nobody retuned it. Cause: full-window layers cost $O(T^2)$ attention, and in a wall-clock frame that converts to tokens through the token law. Rival: those layers carry long-range structure

a hash of recent tokens cannot represent, so removing one costs more than the tokens are worth — which is what removing whole layers did. Separator: the step-time reading, not the endpoint. If attention FLOPs bind, TTTT is measurably faster; if not, it is speed-neutral and any change is pure quality loss.

- Hypothesis. depth $6 \times \{\text{TTTL}, \text{TTTT}, \text{TTLL}\} \times 4$ seeds; both directions measured rather than assuming monotonicity. Anything beating TTTL by ≥ 1 gate goes to $n=10$ through the funnel.
- Reasoning. Against, and this is the honest prior: attention was profiled at $\approx 1.6\%$ of step, capping this lever near -0.001 — under half a gate — unless depth 6 at dim 640 has a materially different attention share. That is why it is a cheap test and not a promising one.
- Ratings. 2 / 3 / 5 / 3 / 4 / 5 / 5.

4.2. E26 — Test the surviving account on its own terms

- Claim. arXiv 2605.29152. Scope recorded as mismatched: BatchNorm ResNets, CIFAR-10, no transformers.
- Mechanism. Observed: refinements are null across duration and dense share. That paper’s cause: initialisation effects are erased by accumulated regularisation, $\mathcal{T}_{\text{adapt}} = \sum \eta_k$ for adaptive optimisers, and it reports the initialisation spread collapsing from 26pp to 0.3pp when L2 is added at fixed duration. Prediction here: conditioned init should show a larger effect when accumulated regularisation is lower. Rival: it is scope-mismatched and predicts nothing for transformers with hashed memories.
- Hypothesis. COND_INIT $\times$ WEIGHT_DECAY $\in \{0, 0.1\}$ at depth 6, $n=4$. At WD=0 the accumulated-regularisation timescale is far smaller, so the account predicts the init effect survives.
- Reasoning. For: it converts an untestable survivor into a falsifiable claim, and WD=0 is free. Against: WD=0 changes the baseline itself, so the contrast is between two non-standard configurations and nothing here can be adopted; and E14 already showed this frame is sensitive to weight decay in ways that may swamp the effect.

- Ratings. 3 / 3 / 4 / 3 / 4 / 5 / 5.

5. Conclusion

We retracted our own explanation of a null, proposed a replacement, and refuted that too. What remains is a well-characterised negative result with no accepted cause: null at 2016 and 3276 steps, alone and stacked, at 3.2% and 34.9% dense share. We think recording that honestly is worth more than the third guess would have been.